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ADJUSTABLE UNIVERSAL CHARGER

Abstract

Methods, systems, and devices for a universal adaptable charger are described. A charger device may include a charging post with a plurality of portions, where a first portion of the plurality of portions is configured to move in a first direction relative to a second portion of the plurality of portions, and where the charging post is configured to receive a wearable ring device. A locking component of the charger device may be configured to prevent the first portion to move relative to the second portion when the charger device is in a locked configuration, and a charging element disposed within one or more of the plurality of portions of the charging post may be configured to charge the wearable ring device through direct or indirect electrical coupling with an additional charging element of the wearable ring device when the wearable ring device is received by the charging post.

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Background/Summary

FIELD OF TECHNOLOGY

[0001] The following relates to wearable devices and data processing, including adjustable universal charger.

BACKGROUND

[0002] Some wearable devices may be configured to collect data from users, including temperature data, heart rate data, and the like. The wearable devices may be configured to charge on a charger base. However, poor connection with a charger base may prevent the wearable device from charging as expected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 illustrates an example of a system that supports an adjustable universal charger in accordance with aspects of the present disclosure.

[0004] FIG. 2 illustrates an example of a system that supports an adjustable universal charger in accordance with aspects of the present disclosure.

[0005] FIGS. 3A and 3B show examples of a charger devices that supports an adjustable universal charger in accordance with aspects of the present disclosure.

[0006] FIG. 4 shows an example of a charger device that supports an adjustable universal charger in accordance with aspects of the present disclosure.

[0007] FIG. 5 shows an example of a movement diagram that supports an adjustable universal charger in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0008] Wearable devices may be configured to collect physiological data from users, such as light-based photoplethysmogram (PPG) data. However, the level or quality of skin contact between the wearable device and the user's tissue may affect the quality of PPG measurements. As such, wearable devices may be manufactured in varying sizes (e.g., circumferential sizes) to achieve good skin contact and a comfortable fit for different users, and to cover respective variations in user fit. In some cases, a wearable device may be associated with a charger manufactured for the size of the respective wearable device. That is, the size of the charger may be specific to the size of the user's wearable device. For this reason, however, it may be difficult for a user to charge a wearable device on a charger corresponding to a smaller size, as the wearable device may exhibit poor contact with the charger device, reducing charging effectiveness (e.g., for wireless inductive charging). These issues may result in the wearable device failing to charge or charging at a relatively slow speed. Further, wearable devices having corresponding charger devices may be inconvenient for households that may have various wearable devices of different sizes, as it may be difficult to share the charger devices between the wearable devices.

[0009] In accordance with examples as described herein, a charger device can be adjustable such as to fit wearable ring devices of varying sizes. For example, the charger device may include a charging post that is divided into two portions, and the two portions may move away from each other when a wearable ring device is positioned around the charging post. As such, each portion of the charging post may press against an inner surface of the wearable ring device, and the universal charging may accommodate wearable ring devices of various sizes for charging. In some examples, the two portions of the charging post may be connected via a spring, which may push the two portions away from each other. In some cases, the universal charging may include a mechanism (e.g., a ratcheting mechanism) which may lock the two portions of the charging post from moving,

and to release the mechanism, the universal charging may include a button. For example, a user may press the button after placing the wearable ring device on the universal charging, thereby allowing the spring to push the two portions of the charging post apart and securing the wearable ring device in place for charging. By using the universal charging, inductive charging components of the universal charging and the wearable ring device may be positioned closely for charging, thereby leading to more effective charging for a wearable device (e.g., faster charging, stronger charge signal, reduced or eliminated charging errors). Further, the charger device may increase user experience by decreasing the quantity of charger devices that a user or household may use to support charging wearable ring devices of different sizes.

[0010] Aspects of the disclosure are initially described in the context of systems supporting physiological data collection from users via wearable devices. Aspects of the disclosure are further described in the context of charger devices and movement diagrams supporting adjustability in accordance with examples as described herein. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to adjustable universal charger.

[0011] FIG. 1 illustrates an example of a system **100** that supports an adjustable universal charger in accordance with aspects of the present disclosure. The system **100** includes a plurality of electronic devices (e.g., wearable devices **104**, user devices **106**) that may be worn and/or operated by one or more users **102**. The system **100** further includes a network **108** and one or more servers **110**.

[0012] The electronic devices may include any electronic devices known in the art, including wearable devices **104** (e.g., ring wearable devices, watch wearable devices, etc.), user devices **106** (e.g., smartphones, laptops, tablets). The electronic devices associated with the respective users **102** may include one or more of the following functionalities: 1) measuring physiological data, 2) storing the measured data, 3) processing the data, 4) providing outputs (e.g., via GUIs) to a user **102** based on the processed data, and 5) communicating data with one another and/or other computing devices. Different electronic devices may perform one or more of the functionalities.

[0013] Example wearable devices **104** may include wearable computing devices, such as a ring computing device (hereinafter “ring”) configured to be worn on a user's **102** finger, a wrist computing device (e.g., a smart watch, fitness band, or bracelet) configured to be worn on a user's **102** wrist, and/or a head mounted computing device (e.g., glasses/goggles). Wearable devices **104** may also include bands, straps (e.g., flexible or inflexible bands or straps), stick-on sensors, and the like, that may be positioned in other locations, such as bands around the head (e.g., a forehead headband), arm (e.g., a forearm band and/or bicep band), and/or leg (e.g., a thigh or calf band), behind the ear, under the armpit, and the like. Wearable devices **104** may also be attached to, or included in, articles of clothing. For example, wearable devices **104** may be included in pockets and/or pouches on clothing. As another example, wearable device **104** may be clipped and/or pinned to clothing, or may otherwise be maintained within the vicinity of the user **102**. Example articles of clothing may include, but are not limited to, hats, shirts, gloves, pants, socks, outerwear (e.g., jackets), and undergarments. In some implementations, wearable devices **104** may be included with other types of devices such as training/sporting devices that are used during physical activity. For example, wearable devices **104** may be attached to, or included in, a bicycle, skis, a tennis racket, a golf club, and/or training weights.

[0014] Much of the present disclosure may be described in the context of a ring wearable device **104**. Accordingly, the terms “ring **104**,” “wearable device **104**,” and like terms, may be used interchangeably, unless noted otherwise herein. However, the use of the term “ring **104**” is not to be regarded as limiting, as it is contemplated herein that aspects of the present disclosure may be performed using other wearable devices (e.g., watch wearable devices, necklace wearable device, bracelet wearable devices, earring wearable devices, anklet wearable devices, and the like).

[0015] In some aspects, user devices **106** may include handheld mobile computing devices, such as

smartphones and tablet computing devices. User devices **106** may also include personal computers, such as laptop and desktop computing devices. Other example user devices **106** may include server computing devices that may communicate with other electronic devices (e.g., via the Internet). In some implementations, computing devices may include medical devices, such as external wearable computing devices (e.g., Holter monitors). Medical devices may also include implantable medical devices, such as pacemakers and cardioverter defibrillators. Other example user devices **106** may include home computing devices, such as internet of things (IoT) devices (e.g., IoT devices), smart televisions, smart speakers, smart displays (e.g., video call displays), hubs (e.g., wireless communication hubs), security systems, smart appliances (e.g., thermostats and refrigerators), and fitness equipment.

[0016] Some electronic devices (e.g., wearable devices **104**, user devices **106**) may measure physiological parameters of respective users **102**, such as photoplethysmography waveforms, continuous skin temperature, a pulse waveform, respiration rate, heart rate, heart rate variability (HRV), actigraphy, galvanic skin response, pulse oximetry, blood oxygen saturation (SpO₂), blood sugar levels (e.g., glucose metrics), and/or other physiological parameters. Some electronic devices that measure physiological parameters may also perform some/all of the calculations described herein. Some electronic devices may not measure physiological parameters, but may perform some/all of the calculations described herein. For example, a ring (e.g., wearable device **104**), mobile device application, or a server computing device may process received physiological data that was measured by other devices.

[0017] In some implementations, a user **102** may operate, or may be associated with, multiple electronic devices, some of which may measure physiological parameters and some of which may process the measured physiological parameters. In some implementations, a user **102** may have a ring (e.g., wearable device **104**) that measures physiological parameters. The user **102** may also have, or be associated with, a user device **106** (e.g., mobile device, smartphone), where the wearable device **104** and the user device **106** are communicatively coupled to one another. In some cases, the user device **106** may receive data from the wearable device **104** and perform some/all of the calculations described herein. In some implementations, the user device **106** may also measure physiological parameters described herein, such as motion/activity parameters.

[0018] For example, as illustrated in FIG. 1, a first user **102-a** (User 1) may operate, or may be associated with, a wearable device **104-a** (e.g., ring **104-a**) and a user device **106-a** that may operate as described herein. In this example, the user device **106-a** associated with user **102-a** may process/store physiological parameters measured by the ring **104-a**. Comparatively, a second user **102-b** (User 2) may be associated with a ring **104-b**, a watch wearable device **104-c** (e.g., watch **104-c**), and a user device **106-b**, where the user device **106-b** associated with user **102-b** may process/store physiological parameters measured by the ring **104-b** and/or the watch **104-c**. Moreover, an nth user **102-n** (User N) may be associated with an arrangement of electronic devices described herein (e.g., ring **104-n**, user device **106-n**). In some aspects, wearable devices **104** (e.g., rings **104**, watches **104**) and other electronic devices may be communicatively coupled to the user devices **106** of the respective users **102** via Bluetooth, Wi-Fi, and other wireless protocols. Moreover, in some cases, the wearable device **104** and the user device **106** may be included within (or make up) the same device. For example, in some cases, the wearable device **104** may be configured to execute an application associated with the wearable device **104**, and may be configured to display data via a GUI.

[0019] In some implementations, the rings **104** (e.g., wearable devices **104**) of the system **100** may be configured to collect physiological data from the respective users **102** based on arterial blood flow within the user's finger. In particular, a ring **104** may utilize one or more light-emitting components, such as LEDs (e.g., red LEDs, green LEDs) that emit light on the palm-side of a user's finger to collect physiological data based on arterial blood flow within the user's finger. In general, the terms light-emitting components, light-emitting elements, and like terms, may include, but are

not limited to, LEDs, micro LEDs, mini LEDs, laser diodes (LDs) (e.g., vertical cavity surface-emitting lasers (VCSELs), and the like.

[0020] In some cases, the system **100** may be configured to collect physiological data from the respective users **102** based on blood flow diffused into a microvascular bed of skin with capillaries and arterioles. For example, the system **100** may collect PPG data based on a measured amount of blood diffused into the microvascular system of capillaries and arterioles. In some implementations, the ring **104** may acquire the physiological data using a combination of both green and red LEDs. The physiological data may include any physiological data known in the art including, but not limited to, temperature data, accelerometer data (e.g., movement/motion data), heart rate data, HRV data, blood oxygen level data, or any combination thereof.

[0021] The use of both green and red LEDs may provide several advantages over other solutions, as red and green LEDs have been found to have their own distinct advantages when acquiring physiological data under different conditions (e.g., light/dark, active/inactive) and via different parts of the body, and the like. For example, green LEDs have been found to exhibit better performance during exercise. Moreover, using multiple LEDs (e.g., green and red LEDs) distributed around the ring **104** has been found to exhibit superior performance as compared to wearable devices that utilize LEDs that are positioned close to one another, such as within a watch wearable device. Furthermore, the blood vessels in the finger (e.g., arteries, capillaries) are more accessible via LEDs as compared to blood vessels in the wrist. In particular, arteries in the wrist are positioned on the bottom of the wrist (e.g., palm-side of the wrist), meaning only capillaries are accessible on the top of the wrist (e.g., back of hand side of the wrist), where wearable watch devices and similar devices are typically worn. As such, utilizing LEDs and other sensors within a ring **104** has been found to exhibit superior performance as compared to wearable devices worn on the wrist, as the ring **104** may have greater access to arteries (as compared to capillaries), thereby resulting in stronger signals and more valuable physiological data.

[0022] The electronic devices of the system **100** (e.g., user devices **106**, wearable devices **104**) may be communicatively coupled to one or more servers **110** via wired or wireless communication protocols. For example, as shown in FIG. 1, the electronic devices (e.g., user devices **106**) may be communicatively coupled to one or more servers **110** via a network **108**. The network **108** may implement transfer control protocol and internet protocol (TCP/IP), such as the Internet, or may implement other network **108** protocols. Network connections between the network **108** and the respective electronic devices may facilitate transport of data via email, web, text messages, mail, or any other appropriate form of interaction within a computer network **108**. For example, in some implementations, the ring **104-a** associated with the first user **102-a** may be communicatively coupled to the user device **106-a**, where the user device **106-a** is communicatively coupled to the servers **110** via the network **108**. In additional or alternative cases, wearable devices **104** (e.g., rings **104**, watches **104**) may be directly communicatively coupled to the network **108**.

[0023] The system **100** may offer an on-demand database service between the user devices **106** and the one or more servers **110**. In some cases, the servers **110** may receive data from the user devices **106** via the network **108**, and may store and analyze the data. Similarly, the servers **110** may provide data to the user devices **106** via the network **108**. In some cases, the servers **110** may be located at one or more data centers. The servers **110** may be used for data storage, management, and processing. In some implementations, the servers **110** may provide a web-based interface to the user device **106** via web browsers.

[0024] In some aspects, the system **100** may detect periods of time that a user **102** is asleep, and classify periods of time that the user **102** is asleep into one or more sleep stages (e.g., sleep stage classification). For example, as shown in FIG. 1, User **102-a** may be associated with a wearable device **104-a** (e.g., ring **104-a**) and a user device **106-a**. In this example, the ring **104-a** may collect physiological data associated with the user **102-a**, including temperature, heart rate, HRV, respiratory rate, and the like. In some aspects, data collected by the ring **104-a** may be input to a

machine learning classifier, where the machine learning classifier is configured to determine periods of time that the user **102-a** is (or was) asleep. Moreover, the machine learning classifier may be configured to classify periods of time into different sleep stages, including an awake sleep stage, a rapid eye movement (REM) sleep stage, a light sleep stage (non-REM (NREM)), and a deep sleep stage (NREM). In some aspects, the classified sleep stages may be displayed to the user **102-a** via a GUI of the user device **106-a**. Sleep stage classification may be used to provide feedback to a user **102-a** regarding the user's sleeping patterns, such as recommended bedtimes, recommended wake-up times, and the like. Moreover, in some implementations, sleep stage classification techniques described herein may be used to calculate scores for the respective user, such as Sleep Scores, Readiness Scores, and the like.

[0025] In some aspects, the system **100** may utilize circadian rhythm-derived features to further improve physiological data collection, data processing procedures, and other techniques described herein. The term circadian rhythm may refer to a natural, internal process that regulates an individual's sleep-wake cycle, that repeats approximately every 24 hours. In this regard, techniques described herein may utilize circadian rhythm adjustment models to improve physiological data collection, analysis, and data processing. For example, a circadian rhythm adjustment model may be input into a machine learning classifier along with physiological data collected from the user **102-a** via the wearable device **104-a**. In this example, the circadian rhythm adjustment model may be configured to “weight,” or adjust, physiological data collected throughout a user's natural, approximately 24-hour circadian rhythm. In some implementations, the system may initially start with a “baseline” circadian rhythm adjustment model, and may modify the baseline model using physiological data collected from each user **102** to generate tailored, individualized circadian rhythm adjustment models that are specific to each respective user **102**.

[0026] In some aspects, the system **100** may utilize other biological rhythms to further improve physiological data collection, analysis, and processing by phase of these other rhythms. For example, if a weekly rhythm is detected within an individual's baseline data, then the model may be configured to adjust “weights” of data by day of the week. Biological rhythms that may require adjustment to the model by this method include: 1) ultradian (faster than a day rhythms, including sleep cycles in a sleep state, and oscillations from less than an hour to several hours periodicity in the measured physiological variables during wake state; 2) circadian rhythms; 3) non-endogenous daily rhythms shown to be imposed on top of circadian rhythms, as in work schedules; 4) weekly rhythms, or other artificial time periodicities exogenously imposed (e.g., in a hypothetical culture with 12 day “weeks,” 12 day rhythms could be used); 5) multi-day ovarian rhythms in women and spermatogenesis rhythms in men; 6) lunar rhythms (relevant for individuals living with low or no artificial lights); and 7) seasonal rhythms.

[0027] The biological rhythms are not always stationary rhythms. For example, many women experience variability in ovarian cycle length across cycles, and ultradian rhythms are not expected to occur at exactly the same time or periodicity across days even within a user. As such, signal processing techniques sufficient to quantify the frequency composition while preserving temporal resolution of these rhythms in physiological data may be used to improve detection of these rhythms, to assign phase of each rhythm to each moment in time measured, and to thereby modify adjustment models and comparisons of time intervals. The biological rhythm-adjustment models and parameters can be added in linear or non-linear combinations as appropriate to more accurately capture the dynamic physiological baselines of an individual or group of individuals.

[0028] In some examples, wearable devices **104** may be manufactured in multiple discrete sizes to accommodate a range of user sizes (e.g., finger sizes). In some cases, however, each respective size for a wearable device **104** may be associated with a corresponding size charger device. As such, charging multiple wearable devices **104** using a same charger device may be difficult, as wearable devices **104** may experience difficulty charging (e.g., experience slow charging or no charging) in a charger device that corresponds to a different size. Further, it may be inconvenient for users **102** to

carry multiple charger devices to charge multiple wearable devices **104** of different sizes.

[0029] Accordingly, in some aspects, the respective devices of the system **100** may support techniques for an adjustable universal charger device for charging the wearable devices **104**. For example, a charger device may include a charging post divided into at least two portions, which may be configured to move apart from each other when a wearable device **104** is placed around the charging post. In some examples, the charger device may include a button release to trigger the portions of the charging post moving away from each other, thereby allowing a user **102** to adjust the charger device. The portions of the charging post may be moved back together by the user **102**, enabling the charger device to accept another wearable device **104** of a different size. As such, a single charger device may be used to charge multiple wearable devices **104** of different sizes.

[0030] It should be appreciated by a person skilled in the art that one or more aspects of the disclosure may be implemented in a system **100** to additionally or alternatively solve other problems than those described above. Furthermore, aspects of the disclosure may provide technical improvements to “conventional” systems or processes as described herein. However, the description and appended drawings only include example technical improvements resulting from implementing aspects of the disclosure, and accordingly do not represent all of the technical improvements provided within the scope of the claims.

[0031] FIG. **2** illustrates an example of a system **200** that supports an adjustable universal charger in accordance with aspects of the present disclosure. The system **200** may implement, or be implemented by, system **100**. In particular, system **200** illustrates an example of a ring **104** (e.g., wearable device **104**), a user device **106**, and a server **110**, as described with reference to FIG. **1**.

[0032] In some aspects, the ring **104** may be configured to be worn around a user's finger, and may determine one or more user physiological parameters when worn around the user's finger. Example measurements and determinations may include, but are not limited to, user skin temperature, pulse waveforms, respiratory rate, heart rate, HRV, blood oxygen levels (SpO₂), blood sugar levels (e.g., glucose metrics), and the like.

[0033] The system **200** further includes a user device **106** (e.g., a smartphone) in communication with the ring **104**. For example, the ring **104** may be in wireless and/or wired communication with the user device **106**. In some implementations, the ring **104** may send measured and processed data (e.g., temperature data, photoplethysmogram (PPG) data, motion/accelerometer data, ring input data, and the like) to the user device **106**. The user device **106** may also send data to the ring **104**, such as ring **104** firmware/configuration updates. The user device **106** may process data. In some implementations, the user device **106** may transmit data to the server **110** for processing and/or storage.

[0034] The ring **104** may include a housing **205** that may include an inner housing **205-a** and an outer housing **205-b**. In some aspects, the housing **205** of the ring **104** may store or otherwise include various components of the ring including, but not limited to, device electronics, a power source (e.g., battery **210**, and/or capacitor), one or more substrates (e.g., printable circuit boards) that interconnect the device electronics and/or power source, and the like. The device electronics may include device modules (e.g., hardware/software), such as: a processing module **230-a**, a memory **215**, a communication module **220-a**, a power module **225**, and the like. The device electronics may also include one or more sensors. Example sensors may include one or more temperature sensors **240**, a PPG sensor assembly (e.g., PPG system **235**), and one or more motion sensors **245**.

[0035] The sensors may include associated modules (not illustrated) configured to communicate with the respective components/modules of the ring **104**, and generate signals associated with the respective sensors. In some aspects, each of the components/modules of the ring **104** may be communicatively coupled to one another via wired or wireless connections. Moreover, the ring **104** may include additional and/or alternative sensors or other components that are configured to collect physiological data from the user, including light sensors (e.g., LEDs), oximeters, and the like.

[0036] The ring **104** shown and described with reference to FIG. 2 is provided solely for illustrative purposes. As such, the ring **104** may include additional or alternative components as those illustrated in FIG. 2. Other rings **104** that provide functionality described herein may be fabricated. For example, rings **104** with fewer components (e.g., sensors) may be fabricated. In a specific example, a ring **104** with a single temperature sensor **240** (or other sensor), a power source, and device electronics configured to read the single temperature sensor **240** (or other sensor) may be fabricated. In another specific example, a temperature sensor **240** (or other sensor) may be attached to a user's finger (e.g., using adhesives, wraps, clamps, spring loaded clamps, etc.). In this case, the sensor may be wired to another computing device, such as a wrist worn computing device that reads the temperature sensor **240** (or other sensor). In other examples, a ring **104** that includes additional sensors and processing functionality may be fabricated.

[0037] The housing **205** may include one or more housing **205** components. The housing **205** may include an outer housing **205-b** component (e.g., a shell) and an inner housing **205-a** component (e.g., a molding). The housing **205** may include additional components (e.g., additional layers) not explicitly illustrated in FIG. 2. For example, in some implementations, the ring **104** may include one or more insulating layers that electrically insulate the device electronics and other conductive materials (e.g., electrical traces) from the outer housing **205-b** (e.g., a metal outer housing **205-b**). The housing **205** may provide structural support for the device electronics, battery **210**, substrate(s), and other components. For example, the housing **205** may protect the device electronics, battery **210**, and substrate(s) from mechanical forces, such as pressure and impacts. The housing **205** may also protect the device electronics, battery **210**, and substrate(s) from water and/or other chemicals.

[0038] The outer housing **205-b** may be fabricated from one or more materials. In some implementations, the outer housing **205-b** may include a metal, such as titanium, that may provide strength and abrasion resistance at a relatively light weight. The outer housing **205-b** may also be fabricated from other materials, such polymers. In some implementations, the outer housing **205-b** may be protective as well as decorative.

[0039] The inner housing **205-a** may be configured to interface with the user's finger. The inner housing **205-a** may be formed from a polymer (e.g., a medical grade polymer) or other material. In some implementations, the inner housing **205-a** may be transparent. For example, the inner housing **205-a** may be transparent to light emitted by the PPG light emitting diodes (LEDs). In some implementations, the inner housing **205-a** component may be molded onto the outer housing **205-b**. For example, the inner housing **205-a** may include a polymer that is molded (e.g., injection molded) to fit into an outer housing **205-b** metallic shell.

[0040] The ring **104** may include one or more substrates (not illustrated). The device electronics and battery **210** may be included on the one or more substrates. For example, the device electronics and battery **210** may be mounted on one or more substrates. Example substrates may include one or more printed circuit boards (PCBs), such as flexible PCB (e.g., polyimide). In some implementations, the electronics/battery **210** may include surface mounted devices (e.g., surface-mount technology (SMT) devices) on a flexible PCB. In some implementations, the one or more substrates (e.g., one or more flexible PCBs) may include electrical traces that provide electrical communication between device electronics. The electrical traces may also connect the battery **210** to the device electronics.

[0041] The device electronics, battery **210**, and substrates may be arranged in the ring **104** in a variety of ways. In some implementations, one substrate that includes device electronics may be mounted along the bottom of the ring **104** (e.g., the bottom half), such that the sensors (e.g., PPG system **235**, temperature sensors **240**, motion sensors **245**, and other sensors) interface with the underside of the user's finger. In these implementations, the battery **210** may be included along the top portion of the ring **104** (e.g., on another substrate).

[0042] The various components/modules of the ring **104** represent functionality (e.g., circuits and

other components) that may be included in the ring **104**. Modules may include any discrete and/or integrated electronic circuit components that implement analog and/or digital circuits capable of producing the functions attributed to the modules herein. For example, the modules may include analog circuits (e.g., amplification circuits, filtering circuits, analog/digital conversion circuits, and/or other signal conditioning circuits). The modules may also include digital circuits (e.g., combinational or sequential logic circuits, memory circuits etc.).

[0043] The memory **215** (memory module) of the ring **104** may include any volatile, non-volatile, magnetic, or electrical media, such as a random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other memory device. The memory **215** may store any of the data described herein. For example, the memory **215** may be configured to store data (e.g., motion data, temperature data, PPG data) collected by the respective sensors and PPG system **235**. Furthermore, memory **215** may include instructions that, when executed by one or more processing circuits, cause the modules to perform various functions attributed to the modules herein. The device electronics of the ring **104** described herein are only example device electronics. As such, the types of electronic components used to implement the device electronics may vary based on design considerations.

[0044] The functions attributed to the modules of the ring **104** described herein may be embodied as one or more processors, hardware, firmware, software, or any combination thereof. Depiction of different features as modules is intended to highlight different functional aspects and does not necessarily imply that such modules must be realized by separate hardware/software components. Rather, functionality associated with one or more modules may be performed by separate hardware/software components or integrated within common hardware/software components.

[0045] The processing module **230-a** of the ring **104** may include one or more processors (e.g., processing units), microcontrollers, digital signal processors, systems on a chip (SOCs), and/or other processing devices. The processing module **230-a** communicates with the modules included in the ring **104**. For example, the processing module **230-a** may transmit/receive data to/from the modules and other components of the ring **104**, such as the sensors. As described herein, the modules may be implemented by various circuit components. Accordingly, the modules may also be referred to as circuits (e.g., a communication circuit and power circuit).

[0046] The processing module **230-a** may communicate with the memory **215**. The memory **215** may include computer-readable instructions that, when executed by the processing module **230-a**, cause the processing module **230-a** to perform the various functions attributed to the processing module **230-a** herein. In some implementations, the processing module **230-a** (e.g., a microcontroller) may include additional features associated with other modules, such as communication functionality provided by the communication module **220-a** (e.g., an integrated Bluetooth Low Energy transceiver) and/or additional onboard memory **215**.

[0047] The communication module **220-a** may include circuits that provide wireless and/or wired communication with the user device **106** (e.g., communication module **220-b** of the user device **106**). In some implementations, the communication modules **220-a**, **220-b** may include wireless communication circuits, such as Bluetooth circuits and/or Wi-Fi circuits. In some implementations, the communication modules **220-a**, **220-b** can include wired communication circuits, such as Universal Serial Bus (USB) communication circuits. Using the communication module **220-a**, the ring **104** and the user device **106** may be configured to communicate with each other. The processing module **230-a** of the ring may be configured to transmit/receive data to/from the user device **106** via the communication module **220-a**. Example data may include, but is not limited to, motion data, temperature data, pulse waveforms, heart rate data, HRV data, PPG data, and status updates (e.g., charging status, battery charge level, and/or ring **104** configuration settings). The processing module **230-a** of the ring may also be configured to receive updates (e.g., software/firmware updates) and data from the user device **106**.

[0048] The ring **104** may include a battery **210** (e.g., a rechargeable battery **210**). An example battery **210** may include a Lithium-Ion or Lithium-Polymer type battery **210**, although a variety of battery **210** options are possible. The battery **210** may be wirelessly charged. In some implementations, the ring **104** may include a power source other than the battery **210**, such as a capacitor. The power source (e.g., battery **210** or capacitor) may have a curved geometry that matches the curve of the ring **104**. In some aspects, a charger or other power source may include additional sensors that may be used to collect data in addition to, or that supplements, data collected by the ring **104** itself. Moreover, a charger or other power source for the ring **104** may function as a user device **106**, in which case the charger or other power source for the ring **104** may be configured to receive data from the ring **104**, store and/or process data received from the ring **104**, and communicate data between the ring **104** and the servers **110**.

[0049] In some aspects, the ring **104** includes a power module **225** that may control charging of the battery **210**. For example, the power module **225** may interface with an external wireless charger that charges the battery **210** when interfaced with the ring **104**. The charger may include a datum structure that mates with a ring **104** datum structure to create a specified orientation with the ring **104** during charging. The power module **225** may also regulate voltage(s) of the device electronics, regulate power output to the device electronics, and monitor the state of charge of the battery **210**. In some implementations, the battery **210** may include a protection circuit module (PCM) that protects the battery **210** from high current discharge, over voltage during charging, and under voltage during discharge. The power module **225** may also include electro-static discharge (ESD) protection.

[0050] The one or more temperature sensors **240** may be electrically coupled to the processing module **230-a**. The temperature sensor **240** may be configured to generate a temperature signal (e.g., temperature data) that indicates a temperature read or sensed by the temperature sensor **240**. The processing module **230-a** may determine a temperature of the user in the location of the temperature sensor **240**. For example, in the ring **104**, temperature data generated by the temperature sensor **240** may indicate a temperature of a user at the user's finger (e.g., skin temperature). In some implementations, the temperature sensor **240** may contact the user's skin. In other implementations, a portion of the housing **205** (e.g., the inner housing **205-a**) may form a barrier (e.g., a thin, thermally conductive barrier) between the temperature sensor **240** and the user's skin. In some implementations, portions of the ring **104** configured to contact the user's finger may have thermally conductive portions and thermally insulative portions. The thermally conductive portions may conduct heat from the user's finger to the temperature sensors **240**. The thermally insulative portions may insulate portions of the ring **104** (e.g., the temperature sensor **240**) from ambient temperature.

[0051] In some implementations, the temperature sensor **240** may generate a digital signal (e.g., temperature data) that the processing module **230-a** may use to determine the temperature. As another example, in cases where the temperature sensor **240** includes a passive sensor, the processing module **230-a** (or a temperature sensor **240** module) may measure a current/voltage generated by the temperature sensor **240** and determine the temperature based on the measured current/voltage. Example temperature sensors **240** may include a thermistor, such as a negative temperature coefficient (NTC) thermistor, or other types of sensors including resistors, transistors, diodes, and/or other electrical/electronic components.

[0052] The processing module **230-a** may sample the user's temperature over time. For example, the processing module **230-a** may sample the user's temperature according to a sampling rate. An example sampling rate may include one sample per second, although the processing module **230-a** may be configured to sample the temperature signal at other sampling rates that are higher or lower than one sample per second. In some implementations, the processing module **230-a** may sample the user's temperature continuously throughout the day and night. Sampling at a sufficient rate (e.g., one sample per second) throughout the day may provide sufficient temperature data for

analysis described herein.

[0053] The processing module **230-a** may store the sampled temperature data in memory **215**. In some implementations, the processing module **230-a** may process the sampled temperature data. For example, the processing module **230-a** may determine average temperature values over a period of time. In one example, the processing module **230-a** may determine an average temperature value each minute by summing all temperature values collected over the minute and dividing by the number of samples over the minute. In a specific example where the temperature is sampled at one sample per second, the average temperature may be a sum of all sampled temperatures for one minute divided by sixty seconds. The memory **215** may store the average temperature values over time. In some implementations, the memory **215** may store average temperatures (e.g., one per minute) instead of sampled temperatures in order to conserve memory **215**.

[0054] The sampling rate, which may be stored in memory **215**, may be configurable. In some implementations, the sampling rate may be the same throughout the day and night. In other implementations, the sampling rate may be changed throughout the day/night. In some implementations, the ring **104** may filter/reject temperature readings, such as large spikes in temperature that are not indicative of physiological changes (e.g., a temperature spike from a hot shower). In some implementations, the ring **104** may filter/reject temperature readings that may not be reliable due to other factors, such as excessive motion during exercise (e.g., as indicated by a motion sensor **245**).

[0055] The ring **104** (e.g., communication module) may transmit the sampled and/or average temperature data to the user device **106** for storage and/or further processing. The user device **106** may transfer the sampled and/or average temperature data to the server **110** for storage and/or further processing.

[0056] Although the ring **104** is illustrated as including a single temperature sensor **240**, the ring **104** may include multiple temperature sensors **240** in one or more locations, such as arranged along the inner housing **205-a** near the user's finger. In some implementations, the temperature sensors **240** may be stand-alone temperature sensors **240**. Additionally, or alternatively, one or more temperature sensors **240** may be included with other components (e.g., packaged with other components), such as with the accelerometer and/or processor.

[0057] The processing module **230-a** may acquire and process data from multiple temperature sensors **240** in a similar manner described with respect to a single temperature sensor **240**. For example, the processing module **230** may individually sample, average, and store temperature data from each of the multiple temperature sensors **240**. In other examples, the processing module **230-a** may sample the sensors at different rates and average/store different values for the different sensors. In some implementations, the processing module **230-a** may be configured to determine a single temperature based on the average of two or more temperatures determined by two or more temperature sensors **240** in different locations on the finger.

[0058] The temperature sensors **240** on the ring **104** may acquire distal temperatures at the user's finger (e.g., any finger). For example, one or more temperature sensors **240** on the ring **104** may acquire a user's temperature from the underside of a finger or at a different location on the finger. In some implementations, the ring **104** may continuously acquire distal temperature (e.g., at a sampling rate). Although distal temperature measured by a ring **104** at the finger is described herein, other devices may measure temperature at the same/different locations. In some cases, the distal temperature measured at a user's finger may differ from the temperature measured at a user's wrist or other external body location. Additionally, the distal temperature measured at a user's finger (e.g., a "shell" temperature) may differ from the user's core temperature. As such, the ring **104** may provide a useful temperature signal that may not be acquired at other internal/external locations of the body. In some cases, continuous temperature measurement at the finger may capture temperature fluctuations (e.g., small or large fluctuations) that may not be evident in core

temperature. For example, continuous temperature measurement at the finger may capture minute-to-minute or hour-to-hour temperature fluctuations that provide additional insight that may not be provided by other temperature measurements elsewhere in the body.

[0059] The ring **104** may include a PPG system **235**. The PPG system **235** may include one or more optical transmitters that transmit light. The PPG system **235** may also include one or more optical receivers that receive light transmitted by the one or more optical transmitters. An optical receiver may generate a signal (hereinafter “PPG” signal) that indicates an amount of light received by the optical receiver. The optical transmitters may illuminate a region of the user's finger. The PPG signal generated by the PPG system **235** may indicate the perfusion of blood in the illuminated region. For example, the PPG signal may indicate blood volume changes in the illuminated region caused by a user's pulse pressure. The processing module **230-a** may sample the PPG signal and determine a user's pulse waveform based on the PPG signal. The processing module **230-a** may determine a variety of physiological parameters based on the user's pulse waveform, such as a user's respiratory rate, heart rate, HRV, oxygen saturation, and other circulatory parameters.

[0060] In some implementations, the PPG system **235** may be configured as a reflective PPG system **235** where the optical receiver(s) receive transmitted light that is reflected through the region of the user's finger. In some implementations, the PPG system **235** may be configured as a transmissive PPG system **235** where the optical transmitter(s) and optical receiver(s) are arranged opposite to one another, such that light is transmitted directly through a portion of the user's finger to the optical receiver(s).

[0061] The number and ratio of transmitters and receivers included in the PPG system **235** may vary. Example optical transmitters may include light-emitting diodes (LEDs). The optical transmitters may transmit light in the infrared spectrum and/or other spectrums. Example optical receivers may include, but are not limited to, photosensors, phototransistors, and photodiodes. The optical receivers may be configured to generate PPG signals in response to the wavelengths received from the optical transmitters. The location of the transmitters and receivers may vary. Additionally, a single device may include reflective and/or transmissive PPG systems **235**.

[0062] The PPG system **235** illustrated in FIG. **2** may include a reflective PPG system **235** in some implementations. In these implementations, the PPG system **235** may include a centrally located optical receiver (e.g., at the bottom of the ring **104**) and two optical transmitters located on each side of the optical receiver. In this implementation, the PPG system **235** (e.g., optical receiver) may generate the PPG signal based on light received from one or both of the optical transmitters. In other implementations, other placements, combinations, and/or configurations of one or more optical transmitters and/or optical receivers are contemplated.

[0063] The processing module **230-a** may control one or both of the optical transmitters to transmit light while sampling the PPG signal generated by the optical receiver. In some implementations, the processing module **230-a** may cause the optical transmitter with the stronger received signal to transmit light while sampling the PPG signal generated by the optical receiver. For example, the selected optical transmitter may continuously emit light while the PPG signal is sampled at a sampling rate (e.g., 250 Hz).

[0064] Sampling the PPG signal generated by the PPG system **235** may result in a pulse waveform that may be referred to as a “PPG.” The pulse waveform may indicate blood pressure vs time for multiple cardiac cycles. The pulse waveform may include peaks that indicate cardiac cycles. Additionally, the pulse waveform may include respiratory induced variations that may be used to determine respiration rate. The processing module **230-a** may store the pulse waveform in memory **215** in some implementations. The processing module **230-a** may process the pulse waveform as it is generated and/or from memory **215** to determine user physiological parameters described herein.

[0065] The processing module **230-a** may determine the user's heart rate based on the pulse waveform. For example, the processing module **230-a** may determine heart rate (e.g., in beats per minute) based on the time between peaks in the pulse waveform. The time between peaks may be

referred to as an interbeat interval (IBI). The processing module **230-a** may store the determined heart rate values and IBI values in memory **215**.

[0066] The processing module **230-a** may determine HRV over time. For example, the processing module **230-a** may determine HRV based on the variation in the IBIs. The processing module **230-a** may store the HRV values over time in the memory **215**. Moreover, the processing module **230-a** may determine the user's respiratory rate over time. For example, the processing module **230-a** may determine respiratory rate based on frequency modulation, amplitude modulation, or baseline modulation of the user's IBI values over a period of time. Respiratory rate may be calculated in breaths per minute or as another breathing rate (e.g., breaths per 30 seconds). The processing module **230-a** may store user respiratory rate values over time in the memory **215**.

[0067] The ring **104** may include one or more motion sensors **245**, such as one or more accelerometers (e.g., 6-D accelerometers) and/or one or more gyroscopes (gyros). The motion sensors **245** may generate motion signals that indicate motion of the sensors. For example, the ring **104** may include one or more accelerometers that generate acceleration signals that indicate acceleration of the accelerometers. As another example, the ring **104** may include one or more gyro sensors that generate gyro signals that indicate angular motion (e.g., angular velocity) and/or changes in orientation. The motion sensors **245** may be included in one or more sensor packages. An example accelerometer/gyro sensor is a Bosch BM1160 inertial micro electro-mechanical system (MEMS) sensor that may measure angular rates and accelerations in three perpendicular axes.

[0068] The processing module **230-a** may sample the motion signals at a sampling rate (e.g., 50 Hz) and determine the motion of the ring **104** based on the sampled motion signals. For example, the processing module **230-a** may sample acceleration signals to determine acceleration of the ring **104**. As another example, the processing module **230-a** may sample a gyro signal to determine angular motion. In some implementations, the processing module **230-a** may store motion data in memory **215**. Motion data may include sampled motion data as well as motion data that is calculated based on the sampled motion signals (e.g., acceleration and angular values).

[0069] The ring **104** may store a variety of data described herein. For example, the ring **104** may store temperature data, such as raw sampled temperature data and calculated temperature data (e.g., average temperatures). As another example, the ring **104** may store PPG signal data, such as pulse waveforms and data calculated based on the pulse waveforms (e.g., heart rate values, IBI values, HRV values, and respiratory rate values). The ring **104** may also store motion data, such as sampled motion data that indicates linear and angular motion.

[0070] The ring **104**, or other computing device, may calculate and store additional values based on the sampled/calculated physiological data. For example, the processing module **230** may calculate and store various metrics, such as sleep metrics (e.g., a Sleep Score), activity metrics, and readiness metrics. In some implementations, additional values/metrics may be referred to as “derived values.” The ring **104**, or other computing/wearable device, may calculate a variety of values/metrics with respect to motion. Example derived values for motion data may include, but are not limited to, motion count values, regularity values, intensity values, metabolic equivalence of task values (METs), and orientation values. Motion counts, regularity values, intensity values, and METs may indicate an amount of user motion (e.g., velocity/acceleration) over time. Orientation values may indicate how the ring **104** is oriented on the user's finger and if the ring **104** is worn on the left hand or right hand.

[0071] In some implementations, motion counts and regularity values may be determined by counting a number of acceleration peaks within one or more periods of time (e.g., one or more 30 second to 1 minute periods). Intensity values may indicate a number of movements and the associated intensity (e.g., acceleration values) of the movements. The intensity values may be categorized as low, medium, and high, depending on associated threshold acceleration values. METs may be determined based on the intensity of movements during a period of time (e.g., 30

seconds), the regularity/irregularity of the movements, and the number of movements associated with the different intensities.

[0072] In some implementations, the processing module **230-a** may compress the data stored in memory **215**. For example, the processing module **230-a** may delete sampled data after making calculations based on the sampled data. As another example, the processing module **230-a** may average data over longer periods of time in order to reduce the number of stored values. In a specific example, if average temperatures for a user over one minute are stored in memory **215**, the processing module **230-a** may calculate average temperatures over a five minute time period for storage, and then subsequently erase the one minute average temperature data. The processing module **230-a** may compress data based on a variety of factors, such as the total amount of used/available memory **215** and/or an elapsed time since the ring **104** last transmitted the data to the user device **106**.

[0073] Although a user's physiological parameters may be measured by sensors included on a ring **104**, other devices may measure a user's physiological parameters. For example, although a user's temperature may be measured by a temperature sensor **240** included in a ring **104**, other devices may measure a user's temperature. In some examples, other wearable devices (e.g., wrist devices) may include sensors that measure user physiological parameters. Additionally, medical devices, such as external medical devices (e.g., wearable medical devices) and/or implantable medical devices, may measure a user's physiological parameters. One or more sensors on any type of computing device may be used to implement the techniques described herein.

[0074] The physiological measurements may be taken continuously throughout the day and/or night. In some implementations, the physiological measurements may be taken during portions of the day and/or portions of the night. In some implementations, the physiological measurements may be taken in response to determining that the user is in a specific state, such as an active state, resting state, and/or a sleeping state. For example, the ring **104** can make physiological measurements in a resting/sleep state in order to acquire cleaner physiological signals. In one example, the ring **104** or other device/system may detect when a user is resting and/or sleeping and acquire physiological parameters (e.g., temperature) for that detected state. The devices/systems may use the resting/sleep physiological data and/or other data when the user is in other states in order to implement the techniques of the present disclosure.

[0075] In some implementations, as described previously herein, the ring **104** may be configured to collect, store, and/or process data, and may transfer any of the data described herein to the user device **106** for storage and/or processing. In some aspects, the user device **106** includes a wearable application **250**, an operating system (OS), a web browser application (e.g., web browser **280**), one or more additional applications, and a GUI **275**. The user device **106** may further include other modules and components, including sensors, audio devices, haptic feedback devices, and the like. The wearable application **250** may include an example of an application (e.g., "app") that may be installed on the user device **106**. The wearable application **250** may be configured to acquire data from the ring **104**, store the acquired data, and process the acquired data as described herein. For example, the wearable application **250** may include a user interface (UI) module **255**, an acquisition module **260**, a processing module **230-b**, a communication module **220-b**, and a storage module (e.g., database **265**) configured to store application data.

[0076] In some cases, the wearable device **104** and the user device **106** may be included within (or make up) the same device. For example, in some cases, the wearable device **104** may be configured to execute the wearable application **250**, and may be configured to display data via the GUI **275**.

[0077] The various data processing operations described herein may be performed by the ring **104**, the user device **106**, the servers **110**, or any combination thereof. For example, in some cases, data collected by the ring **104** may be pre-processed and transmitted to the user device **106**. In this example, the user device **106** may perform some data processing operations on the received data, may transmit the data to the servers **110** for data processing, or both. For instance, in some cases,

the user device **106** may perform processing operations that require relatively low processing power and/or operations that require a relatively low latency, whereas the user device **106** may transmit the data to the servers **110** for processing operations that require relatively high processing power and/or operations that may allow relatively higher latency.

[0078] In some aspects, the ring **104**, user device **106**, and server **110** of the system **200** may be configured to evaluate sleep patterns for a user. In particular, the respective components of the system **200** may be used to collect data from a user via the ring **104**, and generate one or more scores (e.g., Sleep Score, Readiness Score) for the user based on the collected data. For example, as noted previously herein, the ring **104** of the system **200** may be worn by a user to collect data from the user, including temperature, heart rate, HRV, and the like. Data collected by the ring **104** may be used to determine when the user is asleep in order to evaluate the user's sleep for a given “sleep day.” In some aspects, scores may be calculated for the user for each respective sleep day, such that a first sleep day is associated with a first set of scores, and a second sleep day is associated with a second set of scores. Scores may be calculated for each respective sleep day based on data collected by the ring **104** during the respective sleep day. Scores may include, but are not limited to, Sleep Scores, Readiness Scores, and the like.

[0079] In some cases, “sleep days” may align with the traditional calendar days, such that a given sleep day runs from midnight to midnight of the respective calendar day. In other cases, sleep days may be offset relative to calendar days. For example, sleep days may run from 6:00 pm (18:00) of a calendar day until 6:00 pm (18:00) of the subsequent calendar day. In this example, 6:00 pm may serve as a “cut-off time,” where data collected from the user before 6:00 pm is counted for the current sleep day, and data collected from the user after 6:00 pm is counted for the subsequent sleep day. Due to the fact that most individuals sleep the most at night, offsetting sleep days relative to calendar days may enable the system **200** to evaluate sleep patterns for users in such a manner that is consistent with their sleep schedules. In some cases, users may be able to selectively adjust (e.g., via the GUI) a timing of sleep days relative to calendar days so that the sleep days are aligned with the duration of time that the respective users typically sleep.

[0080] In some implementations, each overall score for a user for each respective day (e.g., Sleep Score, Readiness Score) may be determined/calculated based on one or more “contributors,” “factors,” or “contributing factors.” For example, a user's overall Sleep Score may be calculated based on a set of contributors, including: total sleep, efficiency, restfulness, REM sleep, deep sleep, latency, timing, or any combination thereof. The Sleep Score may include any quantity of contributors. The “total sleep” contributor may refer to the sum of all sleep periods of the sleep day. The “efficiency” contributor may reflect the percentage of time spent asleep compared to time spent awake while in bed, and may be calculated using the efficiency average of long sleep periods (e.g., primary sleep period) of the sleep day, weighted by a duration of each sleep period. The “restfulness” contributor may indicate how restful the user's sleep is, and may be calculated using the average of all sleep periods of the sleep day, weighted by a duration of each period. The restfulness contributor may be based on a “wake up count” (e.g., sum of all the wake-ups (when user wakes up) detected during different sleep periods), excessive movement, and a “got up count” (e.g., sum of all the got-ups (when user gets out of bed) detected during the different sleep periods).

[0081] The “REM sleep” contributor may refer to a sum total of REM sleep durations across all sleep periods of the sleep day including REM sleep. Similarly, the “deep sleep” contributor may refer to a sum total of deep sleep durations across all sleep periods of the sleep day including deep sleep. The “latency” contributor may signify how long (e.g., average, median, longest) the user takes to go to sleep, and may be calculated using the average of long sleep periods throughout the sleep day, weighted by a duration of each period and the number of such periods (e.g., consolidation of a given sleep stage or sleep stages may be its own contributor or weight other contributors). Lastly, the “timing” contributor may refer to a relative timing of sleep periods within the sleep day and/or calendar day, and may be calculated using the average of all sleep periods of

the sleep day, weighted by a duration of each period.

[0082] By way of another example, a user's overall Readiness Score may be calculated based on a set of contributors, including: sleep, sleep balance, heart rate, HRV balance, recovery index, temperature, activity, activity balance, or any combination thereof. The Readiness Score may include any quantity of contributors. The "sleep" contributor may refer to the combined Sleep Score of all sleep periods within the sleep day. The "sleep balance" contributor may refer to a cumulative duration of all sleep periods within the sleep day. In particular, sleep balance may indicate to a user whether the sleep that the user has been getting over some duration of time (e.g., the past two weeks) is in balance with the user's needs. Typically, adults need **7-9** hours of sleep a night to stay healthy, alert, and to perform at their best both mentally and physically. However, it is normal to have an occasional night of bad sleep, so the sleep balance contributor takes into account long-term sleep patterns to determine whether each user's sleep needs are being met. The "resting heart rate" contributor may indicate a lowest heart rate from the longest sleep period of the sleep day (e.g., primary sleep period) and/or the lowest heart rate from naps occurring after the primary sleep period.

[0083] Continuing with reference to the "contributors" (e.g., factors, contributing factors) of the Readiness Score, the "HRV balance" contributor may indicate a highest HRV average from the primary sleep period and the naps happening after the primary sleep period. The HRV balance contributor may help users keep track of their recovery status by comparing their HRV trend over a first time period (e.g., two weeks) to an average HRV over some second, longer time period (e.g., three months). The "recovery index" contributor may be calculated based on the longest sleep period. Recovery index measures how long it takes for a user's resting heart rate to stabilize during the night. A sign of a very good recovery is that the user's resting heart rate stabilizes during the first half of the night, at least six hours before the user wakes up, leaving the body time to recover for the next day. The "body temperature" contributor may be calculated based on the longest sleep period (e.g., primary sleep period) or based on a nap happening after the longest sleep period if the user's highest temperature during the nap is at least 0.5° C. higher than the highest temperature during the longest period. In some aspects, the ring may measure a user's body temperature while the user is asleep, and the system **200** may display the user's average temperature relative to the user's baseline temperature. If a user's body temperature is outside of their normal range (e.g., clearly above or below 0.0), the body temperature contributor may be highlighted (e.g., go to a "Pay attention" state) or otherwise generate an alert for the user.

[0084] In some aspects, the system **200** may support techniques for an adjustable universal charger device for charging rings **104**, as described herein. In some examples, a charger device may include a charging post divided into two or more portions, which may be coupled with a string mechanism configured to move the two or more portions apart from each other when a ring **104** is placed around the charging post. In some cases, at least one portion of the charging post may include an inductive charging component configured to charge a ring **104**. As such, when the two or more portions move apart from each other, the two or more portions may contact an inner surface of the ring **104**. This may improve contact between the inductive charging components of the charger device and inductive charging components of the ring **104**, thereby improving charging for the ring **104**. The portions of the charging post may be configured to move back together, enabling the charger device to accept another ring **104** of a same or a different size (e.g., ring sizes). As such, a single charger device may be used to charge multiple wearable devices **104** of different sizes.

[0085] FIG. 3A and FIG. 3B show example of a charger device **300-a** and a charger device **300-b** that support adjustability to accept multiple sizes of wearable devices **104** in accordance with aspects of the present disclosure. The charger device **300-a** illustrates a first configuration, while the charger device **300-b** illustrates second configuration.

[0086] In the first configuration, a portion **305-a** and a portion **305-b** of a charging post of the charger device **300-a** may be positioned against each other (e.g., in contact or in relatively close

proximity). In some cases, the portion **305-a** and the portion **305-b** may be locked in place by a locking component, and the locking component may be released (e.g., by a user **102**) using a button **310**. For example, the button **310** may allow the portion **305-a**, the portion **305-b**, or both, to move in a respective direction (e.g., a direction radially away from a center of the charger device **300-a**, a direction away from the other portion) to reach the second configuration. The button **310** may allow an internal mechanism, such as a spring mechanism, to exert a force on the portion **305-a** and the portion **305-b** to cause the movement.

[0087] The button **310** may be pressed after a wearable device **104** is placed around the charging post, which may cause the portion **305-a** and the portion **305-b** to move in the respective direction, and the portion **305-a** and the portion **305-b** may stop moving after the portion **305-a** and the portion **305-b** are pressed against an inner surface of the wearable device **104**. Releasing the button **310** may engage the locking component, which may prevent the internal mechanism from exerting the force on the portion **305-a** and the portion **305-b**. In some examples, the portion **305-a** and the portion **305-b** may be pushed back together into the first configuration by an external force (e.g., by a user **102**). For instance, the locking mechanism may allow the portion **305-a** and the portion **305-b** to move in a direction towards the other portion (e.g., radially towards the center of the charger device **300-b**), even when the button **310** is not pressed. For example, the locking mechanism may function as a ratcheting mechanism which may allow movement in one direction. Additionally, or alternatively, the locking mechanism may allow the portion **305-a** and the portion **305-b** to move towards one another if the button **310** is pressed.

[0088] A charging element may be disposed within at least one of the portion **305-a** and the portion **305-b** of the charging post. For example, a charging element may be included within the portion **305-a**, and the charging element may be flexible such as to remain attached during movement of the portion **305-a**. Accordingly, a wearable device **104** placed around the charging post of the charger device **300-b** may be charged via the charging element through direct or indirect electrical coupling.

[0089] For example, as the portion **305-a** may be pressed against an inner surface of the wearable device **104**, the charger device **300-b** may improve contact with a charging component of the wearable device **104**. For example, the charger device **300-b** may enable direct electrical coupling to support wired charging. Alternatively, the charging element may be an inductive charging component, and the charger device **300-b** may place the charging element of the charger device within a threshold distance of the charging component (e.g., an inductive charging component) of the wearable device **104** to improve effective charging. In some cases, the threshold distance may be based on parameters of the charging element, the charging component of the wearable device **104**, or both.

[0090] In some examples, one or more of the portions **305-a** and the portions **305-b** may include a magnetic component. The magnetic component may be configured to interact with one or more magnetic components of the wearable device **104** to orient the wearable device **104** in a position that facilitates charging of the wearable device **104**. For example, the wearable device **104** may be oriented in a first radial orientation which may position a charging element (e.g., inductive charging component) of the wearable device **104** in close proximity (e.g., within the threshold distance) to the charging element of the charging post. In some examples, the magnetic component may be positioned within a portion of the charging post that does not include the charging element. For example, the magnetic component may be disposed within the portion **305-b**, which may reduce interference caused by the magnetic component on charging of the wearable device **104**.

[0091] Additionally, or alternatively, one or more of the portion **305-a** and the portion **305-b** may include geometric features that may assist a user **102** to orient the wearable device **104** to support effective charging. For example, the portion **305-a** and the portion **305-b** may include an indentation (e.g., a notch) or a visual indication (e.g., a mark or notch) that may indicate the first radial orientation for the wearable device **104** to support charging of the wearable device **104**. In

some examples, the geometric features may interface with one or more geometric features (e.g., marks, notches) of the wearable device **104** to orient the wearable device **104** in the first radial orientation. For example, the geometric features of the portion **305-a** or the portion **305-b** may be designed to visually align with geometric features of the wearable device **104** to support charging of the wearable device **104**.

[0092] While the charger device **300-a** and the charger device **300-b** are shown to include the portion **305-a** and the portion **305-b**, in some examples, the charger device **300-a** may include additional portions. For example, the charger device **300-a** and the charger device **300-b** may include multiple smaller portions that form the charging post and are each configured to move away from a center of the charging post when the button **310** is pressed. Additionally, or alternatively, the portions may be different sizes from each other.

[0093] Accordingly, the charger device **300-a** and the charger device **300-b** may support charging for wearable devices **104** of multiple sizes, enabling a single charger device to be used for multiple wearable devices **104** having different sizes.

[0094] FIG. **4** shows an example of a charger device **400** that supports an adjustable universal charger in accordance with aspects of the present disclosure. The charger device **400** illustrates an example of an internal mechanism to allow movement for a portion **405-a** and a portion **405-b** of a charging post of the charger device **400**. The charger device **400** may be an example of the charger device **300-a** and the charger device **300-b**, as described with reference to FIGS. **3A** and **3B**. However, different mechanisms may be supported to support the techniques described with reference to FIGS. **3A** and **3B**.

[0095] The charger device **400** may include a button **410** which may release a locking mechanism, allowing the charger device **400** to transition from a locked configuration to an unlocked configuration. The button **410** may be coupled with a spring **415**, which may push the button outwards after being pressed. When the button is pressed, a component **420** may rotate around an axis of the component **420**, causing a tooth of the component **420** to release a set of gears **425-a**. The set of gears **425-a** may include one or more gears, and may be coupled (e.g., via the teeth of the set of gears **425-a**) to a set of gears **425-b**. For example, the button **410** being depressed may allow rotation of the set of gears **425-a**, which may allow the set of gears **425-b** to rotate.

[0096] The set of gears **425-a** may be coupled (e.g., via a small gear of the set of gears **425-a**) with a component **430-a**, which may be coupled with, or be a part of, the portion **405-a**. Similarly, the set of gears **425-b** may be coupled (e.g., via a small gear of the set of gears **425-b**) with a component **430-b**, which may be coupled with, or be a part of, the portion **405-b**. A spring mechanism **435** may exert a mechanical force on the portion **405-a**, the portion **405-b**, or both, to cause the portion **405-a** to move in a direction relative to the portion **405-b**. For example, when the button **410** is depressed, the component **420** may disengage, allowing for movement of the portion **405-a** and the portion **405-b** and rotation of the set of gears **425-a** and the set of gears **425-b** in one direction (e.g., clockwise for the set of gears **425-a** and counterclockwise for the set of gears **425-b**, as seen from above). The spring mechanism **435** and the associated actions caused by depression of the button **410** are described in more detail with reference to FIG. **5**. Accordingly, the portion **405-a** and the portion **405-b** may move until reaching a final position, or until contacting an inner surface of a wearable device **104**, thereby supporting charging for wearable devices **104** of multiple different sizes.

[0097] When the button **410** is released, the component **420** may engage with the set of gears **425-a** to prevent rotation of the set of gears **425-a** and the set of gears **425-b**, as well as movement of the portion **405-a** and the portion **405-b** in a direction away from the other portion (e.g., away from a center for the charger device **400**). In some examples, the component **420** may allow rotation of the set of gears **425-a** in one direction (e.g., in a counterclockwise direction as seen from above) when the charger device is in the locked configuration. As such, an external force (e.g., by a user **102**) may push the portion **405-a** and the portion **405-b** back together at the center of the charger device

400 without pressing the button **410**.

[0098] The portion **405-a** may include a charging element **440** to support charging of a wearable device **104**. The charging element **440** may be positioned at least partially within the portion **405-a**. In some examples, to support movement of the portion **405-a**, the charging element **440** may be flexible to maintain a connection between the charging element **440** and circuitry of the charger device **400** (e.g., a power supply) while the portion **405-a** moves. For example, the charging element may include an S-shaped connector, a transmitter coil, ferrite tape, or a combination thereof. As described herein with reference to FIGS. **3A** and **3B**, the charging element **440** may be an example of an inductive charging element, which may interact with one or more inductive charging components of a wearable device **104** to support wireless charging of the wearable device **104** when the wearable device **104** is placed on the charger device **400**.

[0099] The charger device **400** may include a PCB **445** (e.g., circuitry), which may be configured to enable charging of wearable devices **104**. For example, the PCB **445** may supply power to the charging element **440**. In some examples, the PCB **445** may receive power from an external power supply via a port (not shown). The PCB **445** may then supply the power internally to the charging element **440**. Additionally, or alternatively, the charger device **400** may include a battery which may be charged by the external power supply, and the battery may supply power to the charging element **440** even when the external power supply is disconnected.

[0100] Accordingly, the charger device **400** may support charging, such as wireless charging, of wearable devices **104** of multiple distinct sizes.

[0101] FIG. **5** shows an example of a movement diagram **500** that supports an adjustable universal charger in accordance with aspects of the present disclosure. The movement diagram **500** illustrates the movement of various components of a charger device, such as the charger device **400** described with reference to FIG. **4**, via a set of arrows. For example, the movement diagram **500** may illustrate how a button **510** may be pressed to cause movement of a portion **505-a** and a portion **505-b** of a charging post in respective directions (e.g., depicted by the arrows).

[0102] When the button **510** is in a released state (e.g., held by a spring component, as described with reference to FIG. **4**), a locking component **520** may engage with a set of gears **525-a**, preventing movement of the set of gears **525-a** in at least a first rotational direction (e.g., illustrated by the respective arrow). As such, a set of gears **525-b** coupled with the set of gears **525-a** may also be prevented from movement in at least a second rotational direction (e.g., illustrated by the respective arrow). The set of gears **525-a** and the set of gears **525-b** may engage with a component **530-a** and a component **530-b**, respectively. The component **530-a** and the component **530-b** may be coupled with or be a part of the portion **505-a** and the portion **505-b**, respectively. As such, when the locking component **520** is engaged and the charger device is in a locked configuration, the movement of the portion **505-a** and the portion **505-b** in the respective directions may be prevented via the set of gears **525-a** and the set of gears **525-b** and via the component **530-a** and the component **530-b**.

[0103] When the button **510** is pressed (e.g., by a user), the portion **505-a** and the portion **505-b** may move in the respective directions due to an external force exerted by a spring mechanism **515**. For example, when the button **510** is pressed, the locking component **520** may disengage from the set of gears **525-a** (e.g., by rotation around an axis of the locking component **520**), allowing for rotation of the set of gears **525-a** in the first rotational direction and movement of the set of gears **525-b** in the second rotational direction. Disengagement of the locking component **520** may also allow for movement of the component **530-a** and the component **530-b** in the respective directions. As such, force exerted by the spring mechanism **515** on the portion **505-a** and the portion **505-b** (e.g., or the component **530-a** and the component **530-b**) may cause the portion **505-a** and the portion **505-b** to move in the respective directions (e.g., as shown by the arrows).

[0104] Accordingly, when the button **510** is pressed, the portion **505-a** and the portion **505-b** may move in the respective directions, and the portions **505-a** and the portions **505-b** may stop moving

after contacting an inner surface of a wearable device **104**. As such, charging of wearable devices **104** of multiple varying sizes by the charger device may be supported, as described herein.

[0105] In some examples, when the button **510** is released and the locking component **520** is engaged, movement of the portion **505-a** and the portion **505-b** may be supported in directions opposite to the respective directions (e.g., opposite to the arrows). For example, the locking component **520** may allow the set of gears **525-a** to rotate in the second rotational direction (e.g., opposite to the first rotational direction shown by the respective arrow). Similarly, the set of gears **525-b** may move in the first rotational direction (e.g., opposite to the second rotational direction shown by the respective arrow). Accordingly, an external force exerted on the portion **505-a** and the portion **505-b** may cause the portion **505-a** and the portion **505-b** to move in the directions opposite to the respective directions, and the portion **505-a** and the portion **505-b** may remain in place after the external force is removed due to the locking component **520** being engaged. This may allow for resetting of the charger device to a configuration in which the charging post is at a smallest size, allowing for a wearable device **104** of a smallest size (e.g., of a set of possible sizes) to fit around the charging post for charging.

[0106] It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

[0107] A charger device is described. The charger device may include a charging post comprising a plurality of portions, wherein a first portion of the plurality of portions is configured to move in a first direction relative to a second portion of the plurality of portions, and wherein the charging post is configured to receive a wearable ring device, a locking component configured to prevent the first portion to move relative to the second portion when the charger device is in a locked configuration, and a charging element disposed within one or more of the plurality of portions of the charging post and configured to charge the wearable ring device through direct or indirect electrical coupling with an additional charging element of the wearable ring device when the wearable ring device is received by the charging post.

[0108] In some examples of the charger device, the first portion may be configured to stop moving in the first direction relative to the second portion based at least in part on the first portion and the second portion being against an inner surface of the wearable ring device.

[0109] In some examples of the charger device, the charging post may be configured to receive wearable ring devices associated with a plurality of sizes and the first portion and the second portion may be configured to position against an inner surface of the wearable ring devices to support charging of the wearable ring devices.

[0110] Some examples of the charger device may further include a spring mechanism configured to exert a mechanical force on at least the first portion to move the first portion in the first direction relative to the second portion when the locking component may be released, wherein the first portion and the second portion may be positioned against the inner surface of the wearable ring devices based at least in part on the mechanical force being exerted on at least the first portion.

[0111] In some examples of the charger device, the locking component may be coupled with the spring mechanism to prevent the spring mechanism from exerting the mechanical force on at least the first portion when the charger device may be in the locked configuration.

[0112] Some examples of the charger device may further include a button configured to release the locking component when the button may be pressed, shifting the charger device from the locked configuration to an unlocked configuration, wherein the first portion moves in the first direction away from the second portion based at least in part on the locking component being released.

[0113] In some examples of the charger device, the button may be further configured to engage the locking component when the button may be released, shifting the charger device from the unlocked configuration to the locked configuration.

[0114] Some examples of the charger device may further include a first set of one or more gears coupled with the first portion of the plurality of portions and a second set of one or more gears coupled with the second portion of the plurality of portions and the first set of one or more gears, wherein, to prevent the first portion to move relative to the second portion, the locking component may be configured to prevent the first set of one or more gears, the second set of one or more gears, or both, from rotating in a first rotational direction.

[0115] In some examples of the charger device, the locking component may be configured to allow the first set of one or more gears, the second set of one or more gears, or both, to rotate in a second rotational direction opposite to the first rotational direction when the charger device may be in the locked configuration based at least in part on an external force exerted on the charging post and the first portion may be configured to move in a second direction opposite to the first direction and towards the second portion in response to the external force.

[0116] In some examples of the charger device, one or more of the plurality of portions of the charging post comprise one or more geometric features that may be configured to interface with one or more additional geometric features of the wearable ring device to orient the wearable ring device in a first radial orientation.

[0117] In some examples of the charger device, the charging element may be configured to charge the wearable ring device based at least in part on the wearable ring device being oriented in the first radial orientation.

[0118] In some examples of the charger device, the plurality of portions of the charging post may be configured to place the charging element within a threshold distance of the additional charging element of the wearable ring device to charge the wearable ring device. 'In some examples of the charger device, the threshold distance may be based at least in part on one or more parameters associated with the charging element, the additional charging element of the wearable ring device, or both.

[0119] Some examples of the charger device may further include a magnetic component disposed within one or more of the plurality of portions of the charging post and configured to interact with one or more magnetic components of the wearable ring device to orient the wearable ring device in a first radial orientation for charging of the wearable ring device. In some examples of the charger device, the charging element comprises a transmitter coil, ferrite tape, or both.

[0120] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term "exemplary" used herein means "serving as an example, instance, or illustration," and not "preferred" or "advantageous over other examples." The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0121] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0122] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0123] The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an

FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0124] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0125] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable ROM (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

[0126] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A charger device, comprising: a charging post comprising a plurality of portions, wherein a first portion of the plurality of portions is configured to move in a first direction relative to a second portion of the plurality of portions, and wherein the charging post is configured to receive a wearable ring device; a locking component configured to prevent the first portion to move relative to the second portion when the charger device is in a locked configuration; and a charging element disposed within one or more of the plurality of portions of the charging post and configured to charge the wearable ring device through direct or indirect electrical coupling with an additional charging element of the wearable ring device when the wearable ring device is received by the charging post.
2. The charger device of claim 1, wherein the first portion is configured to stop moving in the first direction relative to the second portion based at least in part on the first portion and the second portion being against an inner surface of the wearable ring device.
3. The charger device of claim 1, wherein the charging post is configured to receive wearable ring devices associated with a plurality of sizes, and wherein the first portion and the second portion are configured to position against an inner surface of the wearable ring devices to support charging of the wearable ring devices.
4. The charger device of claim 3, further comprising: a spring mechanism configured to exert a mechanical force on at least the first portion to move the first portion in the first direction relative to the second portion when the locking component is released, wherein the first portion and the second portion are positioned against the inner surface of the wearable ring devices based at least in part on the mechanical force being exerted on at least the first portion.
5. The charger device of claim 4, wherein the locking component is coupled with the spring mechanism to prevent the spring mechanism from exerting the mechanical force on at least the first portion when the charger device is in the locked configuration.
6. The charger device of claim 1, further comprising: a button configured to release the locking component when the button is pressed, shifting the charger device from the locked configuration to an unlocked configuration, wherein the first portion moves in the first direction away from the second portion based at least in part on the locking component being released.
7. The charger device of claim 6, wherein the button is further configured to engage the locking component when the button is released, shifting the charger device from the unlocked configuration to the locked configuration.
8. The charger device of claim 1, further comprising: a first set of one or more gears coupled with the first portion of the plurality of portions; and a second set of one or more gears coupled with the second portion of the plurality of portions and the first set of one or more gears, wherein, to prevent the first portion to move relative to the second portion, the locking component is configured to prevent the first set of one or more gears, the second set of one or more gears, or both, from rotating in a first rotational direction.
9. The charger device of claim 8, wherein the locking component is configured to allow the first set of one or more gears, the second set of one or more gears, or both, to rotate in a second rotational direction opposite to the first rotational direction when the charger device is in the locked configuration based at least in part on an external force exerted on the charging post, and wherein the first portion is configured to move in a second direction opposite to the first direction and towards the second portion in response to the external force.
10. The charger device of claim 1, wherein one or more of the plurality of portions of the charging post comprise one or more geometric features that are configured to interface with one or more additional geometric features of the wearable ring device to orient the wearable ring device in a first radial orientation.

11. The charger device of claim 10, wherein the charging element is configured to charge the wearable ring device based at least in part on the wearable ring device being oriented in the first radial orientation.

12. The charger device of claim 1, wherein the plurality of portions of the charging post are configured to place the charging element within a threshold distance of the additional charging element of the wearable ring device to charge the wearable ring device.

13. The charger device of claim 12, wherein the threshold distance is based at least in part on one or more parameters associated with the charging element, the additional charging element of the wearable ring device, or both.

14. The charger device of claim 1, further comprising: a magnetic component disposed within one or more of the plurality of portions of the charging post and configured to interact with one or more magnetic components of the wearable ring device to orient the wearable ring device in a first radial orientation for charging of the wearable ring device.

15. The charger device of claim 1, wherein the charging element comprises a transmitter coil, ferrite tape, or both.
